

"Wings I Can Feel": Haptic Design Principles for Embodied VR Flying

Ruochen Hu
Academy of Arts & Design
Tsinghua University
Beijing, China
hrc23@mails.tsinghua.edu.cn

Xiuge Zhang
Academy of Arts & Design
Tsinghua University
Beijing, China
zhangxg23@mails.tsinghua.edu.cn

Zihui Chen
Tsinghua University
Beijing, China
chen-zh23@mails.tsinghua.edu.cn

Mengshi Yang
College of Design and Innovation
Tongji University
Shanghai, China
yangmengshi07@gmail.com

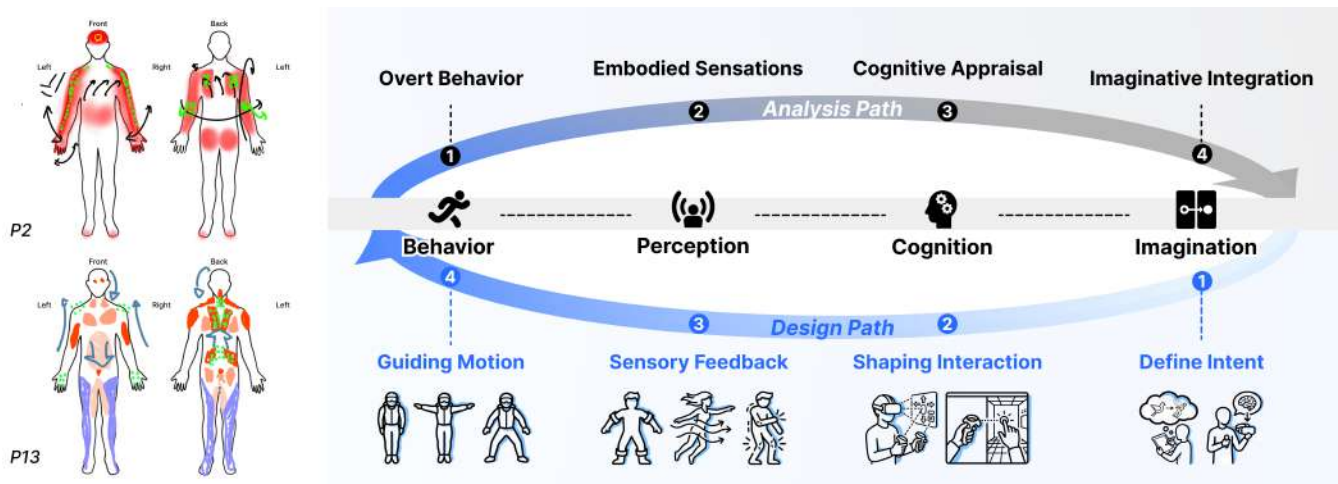


Figure 1: Body map examples and an analysis–design loop for understanding and designing VR flying experiences.

Abstract

VR flying can evoke strong self-motion and freedom, yet many experiences remain visually driven and provide limited bodily evidence. Haptics are often used as generic effects, leaving little flight-specific grounding for interaction design. To investigate flight-specific haptic cues that shape perceived body ownership, we conducted a user study with 14 participants and derived four themes through reflexive thematic analysis. We identify four themes: Bodily Engagement, Multisensory Needs, Control Effectiveness, and Phantom Limb Imagery. We also present a four-layer model that supports both an analysis pathway and a reverse design pathway for deriving design principles. Our findings guide future prototypes and studies and serve as practical lenses for evaluating VR flying experiences.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Haptic Interaction, Virtual Reality, Embodiment, Body Ownership, VR Flying

ACM Reference Format:

Ruochen Hu, Zihui Chen, Xiuge Zhang, and Mengshi Yang. 2026. "Wings I Can Feel": Haptic Design Principles for Embodied VR Flying. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3772363.3798624>

1 Introduction

VR flying can evoke strong self-motion sensations and a sense of freedom [34, 38, 44], making it a compelling embodied experience [24]. Yet many VR flying experiences remain visually driven [26, 32]. When the body lacks corresponding haptic and external-force cues, users may experience action–sensation mismatch, reduced perceived control reliability, and discomfort [29]. Haptic cues that



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798624>

convey aerodynamics and self-motion can provide bodily evidence that grounds the experience [10, 41]. Thus, understanding which bodily cues users rely on is essential for designing effective haptic interactions.

Prior embodiment and body ownership work highlights the role of multisensory congruency and action–feedback coupling [11, 21, 35]. The HCI / XR community has explored diverse haptic modalities, including wearable feedback [6, 18], force [4, 31, 37], and environmental airflow [16, 28, 36], to support immersion and self-motion [16]. However, in VR flying, haptics are often treated as generic effects rather than situated evidence tied to flight control and aerodynamic cues, leaving limited flight-specific grounding for design.

Accordingly, we ask: **RQ1. What flight-specific haptic cues influence users' perceived body ownership during VR flying?** **RQ2. How can these cues be translated into design principles for embodied VR flying?** We conducted a user study with 14 participants combining a VR flying experience, body maps, and semi-structured interviews, and analyzed the data through reflexive thematic analysis to identify needs and opportunities for haptic interaction design. Our study makes the following contributions: (1) we identified a four-layer structure of flight-specific haptic cues and experience factors that shape perceived body ownership during VR flying, and (2) we derived actionable design principles that can inform designing and evaluating embodied VR flying.

2 Related Work

2.1 Embodiment and Body Ownership in VR

In VR, embodiment is often discussed as a multi-component experience involving self-location, agency, and body ownership [22]. Body ownership refers to the feeling that a virtual body (or body part) is one's own [40], and it is often treated as a key dimension for assessing embodiment in VR [14, 22, 25]. Body ownership relies on multisensory congruency. Congruent visual, haptic, and visuomotor cues provide consistent bodily evidence that supports a coherent sense of one's body in VR [9, 11, 39]. Changes in body ownership can also be associated with affect and comfort during an experience [12]. For example, Slater et al. reported that stronger ownership over a virtual body elicited stronger physiological responses when that body was threatened [40]. Moreover, Choi et al. showed that manipulating avatar motion synchrony changed both body ownership and VR sickness [5]. These findings motivate haptic cues as a practical design lever for shaping body ownership, while also suggesting that which cues matter most is likely situated, depending on the activity's control demands and sensory contingencies [7], especially in self-motion-heavy activities (e.g., flying).

However, in non-human flight tasks, ownership may not be directed toward the avatar as a whole. Building on distinctions between avatar-level ownership [40], limb or extension ownership [11, 21], and altered body schema [8, 13], VR flying may engage these levels in combination. Users may experience ownership over flight-relevant body regions, such as the arms or back as control surfaces, non-human appendages, such as wings, or a reconfigured

body schema accommodating non-human morphology. This layered perspective motivates investigating which haptic cues support ownership at each level in VR flying.

2.2 Haptic Cues for Embodied Self-Motion in VR Flying

HCI/XR research has explored diverse haptic modalities for VR, including wearable vibrotactile [6, 19, 28] and pressure-based feedback [18, 31], force [4, 23, 37], thermal feedback [15, 27, 30, 43], and environmental airflow [16, 17] to support embodied interaction. However, much of this work emphasizes device capability and rendering fidelity. As a result, fewer studies examine which cues users rely on, particularly for VR flying, where aerodynamic cues and wing-related control are central. In VR self-motion contexts, airflow cues [16, 33] and vibration [19, 28, 34] are often used as proxy cues to reinforce perceived speed and acceleration alongside visual motion cues. For instance, Shimato et al. reported that wind stimulation increased immersion and the sensation of flying, and noted that perceived self-motion speed may be further shaped through visual motion design such as dynamic background imagery [36]. Commercial and research prototypes tend to be visually led, offering limited flight-specific grounding for designing haptics that match how bodily control and sensations are formed during flight. Motivated by this gap, we investigate flight-specific haptic factors and translate them into design principles for embodied VR flying.

3 Method and Findings

3.1 Data collection

To explore which haptic cues support body ownership in VR flight, we conducted a user study consisting of a VR flight experience, body maps drawing, and semi-structured interviews. The procedure is shown in Figure 2a, and the study materials are provided in Appendix A.2. We recruited 14 participants ($M = 24.79$, $SD = 1.58$; 7 male, 7 female) via social media. All participants self-reported a Chinese cultural background and had prior VR gaming experience. The study was conducted one-on-one in a quiet laboratory setting with a researcher and a note-taker. All participants provided informed consent.

We used *Rush 2*, a commercial VR game, as the VR flight environment because it supports motion-based flight control and provides a stable experience. In *Rush 2*, players control their flight from a first-person perspective by moving their arms to simulate wingsuit flight. Participants played for 20 minutes and were asked to attend to their bodily sensations. During the 20-minute session, participants were instructed to engage their whole body in the experience. First, they completed a fixed-flight course involving maneuvers such as diving, ascending, and changing direction, learning how to control their body for flight. Following this, they performed free-flight in a canyon map, allowing for more dynamic movements. Throughout the experience, participants were asked to focus on their bodily sensations. The freedom to adjust their movements based on the environmental cues within the game helped them find a comfortable state. Following the gameplay session, participants drew body maps to depict: (1) sensation intensity, (2) motion directions, and (3) body areas used for flight control, and then explained their markings. We then conducted semi-structured interviews, first eliciting flight

experiences and bodily sensations and then discussing expectations for the experience. Interviews were audio recorded with participants' consent. All participants received a gift voucher worth 100 RMB (approximately 14 USD) as a token of appreciation.

3.2 Data analysis

We analyzed the interview data using a mixed approach in which thematic analysis [2, 3] was primary and computational methods served for triangulation and coverage checks. We de-identified transcripts and segmented them into meaning units. The first author then conducted iterative open coding and theme development, grouping related codes and refining theme definitions and boundaries across multiple passes. Subsequently, the second author independently reviewed a subset of coded excerpts and the evolving theme structure. Disagreements were resolved through discussion, with all authors adjudicating the final theme names and definitions. In parallel, we applied Latent Dirichlet Allocation (LDA) [1, 20] and NVivo (a computer-assisted qualitative data analysis software) to surface candidate topics and excerpts. Outputs were treated as prompts rather than themes and were incorporated only after manual review. We documented align/merge/split/reject decisions based on semantic fit, boundary clarity, and contribution, and supported trustworthiness through triangulation and negative-case checking. We treated body maps as participant-generated elicitation artifacts, participants' walkthroughs of their maps were audio-recorded, transcribed, and analyzed alongside the interview transcripts, and the drawings were used to contextualize interpretation rather than as quantitative measures of intensity.

3.3 Findings

Based on thematic analysis informed by body maps, we identified four themes which capture the coupling between participants' actions and perception and how they linked these factors to perceived body ownership (Figure 2b).

3.3.1 Bodily Engagement. Participants characterized VR flight control as primarily upper-body-driven, centered on coordinated engagement of the back musculature, core strength, and shoulders/arms. In their body maps (Figure 3), P2, P3, and P4 highlighted these regions with dense markings, indicating where control and effort were most salient. As P3 explained, *"I mainly control it with arms, back, and upper body"*. Participants described using this coordinated engagement to modulate posture for maneuvers such as dives and pull-ups while maintaining balance. Several participants also externalized directional adjustment strategies: for example, P7 and P14 drew arrow-based coordinate systems to illustrate how they corrected posture during motion. This sequencing of control was echoed in accounts such as *"When turning, I first adjust the direction of arms, and then shoulders follow"* (P9) and *"To avoid falling, my core and arms keep exerting force"* (P11).

3.3.2 Multisensory Needs. Participants emphasized that convincing flight required multisensory cues beyond vision, spanning kinesthesia-related cues, air resistance, cutaneous cues, and vestibular sensation. At a kinesthetic level, participants expected flight to be legible through cues that supported self-motion awareness and control, such as wind-related guidance and a clearer sense of

speed and acceleration, rather than being conveyed only through visual effects. In parallel, they described cutaneous cues as essential for anchoring the experience on the body. Rather than perceiving "wind" as a purely visual effect, they wanted airflow to register as tactile sensation on the skin, as P3 put it, *"I crave feeling wind flowing across my body"*. Participants also characterized air resistance as a form of drag or environmental pressure acting on the body during flight, shaping movement and effort; P13 highlighted the corresponding regions in their body map, adding, *"I want my back to experience stronger drag"*. Beyond these cues, several participants reported vestibular-like sensations such as weightlessness and dizziness (e.g., P6) and associated moments of discomfort with a perceived mismatch between what they did and what their body "felt", suggesting that missing or weak haptic cues made action and bodily sensation feel misaligned (P11, P14).

3.3.3 Control Effectiveness. We found that participants experienced control as more effective when action was tightly coupled with bodily sensation. They repeatedly described effective control as emerging from two intertwined qualities: responsiveness, where visual and haptic feedback kept pace with postural adjustments, and a sense of freedom, where flight did not feel overly constrained by the environment or by one's physical capability. Under these conditions, participants reported making fewer corrective movements and shifting attention from actively "managing" control to attending to bodily changes. As P9 recalled, *"I can focus more on bodily changes instead of constantly adjusting my actions"*. Some further suggested that stronger impact cues could intensify this embodied engagement by making speed and motion more legible to the body.

3.3.4 Phantom limbs imagery. Across participants' aspirational flight visions, a common pattern was the desire to achieve flight by extending the body or enhancing bodily capabilities, motivating our theme of phantom limb imagery. Phantom limb imagery often originates from real-world creatures or cinematic depictions, such as bird wings, wing membrane, and flying carpet. Most of the participants imagined bird wings attached around the scapula or back. P3 expressed a desire to *"feel airflow with feathers"*. Others described the wing membrane as a continuous surface that connects the arms and torso. Fewer participants mentioned lower-limb driven, such as flying carpets or hoverboards.

4 Discussion

We derived a four-layer model from thematic analysis to explain how VR flight experiences unfold from control behavior to bodily sensations, cognitive appraisal, and imagined bodily extensions. We use it as both an analysis pathway (overt behavior, embodied sensations, cognitive appraisal, and imaginative integration) and a reverse design pathway for deriving design principles. These principles can also serve as practical evaluation lenses for future VR flying systems, helping assess whether experience supports embodied control and perceived body ownership. Our findings highlight that body ownership in flight experiences is primarily shaped by control over the upper limbs, including the arms, back musculature, and core strength.

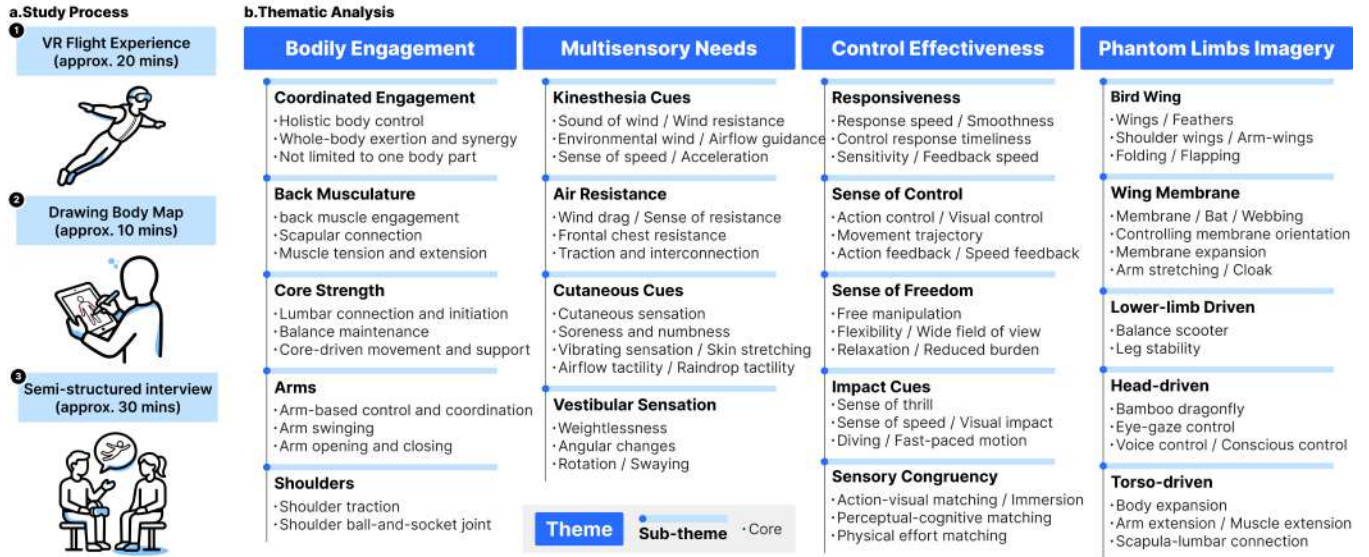


Figure 2: a. Study procedure with three steps and approximate durations—VR flying experience, drawing a body map, and semi-structured interview. b. thematic analysis summary organized into four themes (Bodily Engagement, Multisensory Needs, Control Effectiveness, Phantom Limb Imagery) with associated sub-themes and example items listed under each.

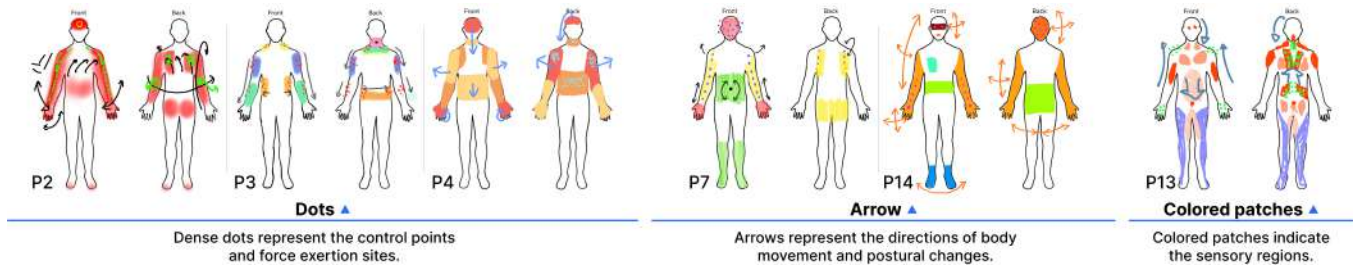


Figure 3: A set of body maps drawn by participants to depict perceived bodily sensations during the task. Each participant is shown with front and back body outlines. Three types of visual annotations are used: dense dots marking locations of control points and force exertion, arrows indicating directions of body movement and postural change, and colored patches highlighting sensory regions.

4.1 From Behavior to Imagination: Empirical Insights Beyond Prior Work

Our findings suggest that VR flying is not a generic full-body embodiment experience. Instead, control is coordinated through the arms and shoulders, back musculature, and core strength. Kilteni et al.'s embodiment framework primarily characterizes embodiment through its core dimensions [22], whereas we add an activity-specific account of bodily role division and control strategies that grounds body ownership in what the body does. This implies that haptic and interaction design should not be anchored only on the hands or head, but should prioritize the scapular and back musculature as a flight-relevant bodily anchor.

Needs for kinesthetic cues, air resistance, and cutaneous cues are not primarily about realism, but about making flight legible at the bodily level. Prior work leverages airflow to enhance immersion and flying sensation, for example, Han et al. used environmental

airflow [16], and Shimato et al. applied directed wind stimuli in VR flying [36]. We extend this line by framing air resistance, such as airflow and drag, as external-force and boundary cues that support control and body ownership through action-sensation coupling. This complements Riecke's work on multisensory cues forvection [34] by situating these cues in an activity context, where missing wind or drag evidence was associated with action-sensation mismatch, discomfort, and weaker body ownership.

Control effectiveness is shaped by responsiveness and control freedom. Choi et al. showed that manipulating avatar motion synchrony changed both body ownership and VR sickness [5]. Building on this, we observed that synchrony was often interpreted as control reliability and closely tied to comfort in this activity, suggesting that low latency and cue consistency should be prioritized before adding complex effects. Finally, participants often entered the experience with phantom limb imagery, such as wings, membranes, vehicles, or head- and intention-driven control. In non-human flight

tasks, such imagery can shape how body ownership is constructed, which may explain why the same haptic cues do not benefit all users equally.

4.2 Design Principles for Embodied VR Flying

Start with the Intended Flying Body. Design should start by clarifying users' intended flying body, such as wings, membranes, vehicles, or head- and intention-driven control. This intent can be set or co-created during onboarding (e.g., selecting attachment points and coverage), and then used to derive control mappings and haptic placement, prioritizing the scapular and back musculature as the primary bodily anchor. In contrast to ownership work that primarily relies on synchrony and avatar substitution [11, 40], we treat the flying body model as an upstream design variable that shapes how body ownership is constructed in this activity.

Responsiveness First, Freedom as Explainable Constraints. Responsiveness should be treated as a minimum design baseline. Key posture changes (e.g., roll, pitch, and yaw) should be reflected through consistent and timely visual and haptic updates, avoiding lag or discontinuities that undermine perceived control reliability. Synchrony should be treated as a baseline requirement: beyond supporting perceived control reliability, timing coherence has been linked to both embodied presence and comfort. For example, Yang and Hu highlight haptic–visual synchrony as a key determinant of embodied presence[42], and Choi et al. proposed that motion synchrony can reduce VR sickness[5]. Control freedom can be supported through explainable constraints rather than opaque limits, for example, using drag-like resistance or damping to make speed changes understandable while preserving controllability.

Moreover, when designing haptic feedback for VR flying, the weight and load of the haptic hardware must be considered, particularly for tasks involving large-body movements. Haptic devices should be lightweight and ergonomically designed to support natural body motion without impeding flight control, ensuring fluid movement without burdening the user with excessive weight or cumbersome equipment.

Make Aerodynamic Cues Legible to the Body. Air resistance should be designed as aerodynamic information rather than decorative effects. Feedback should make direction, intensity, and external-force location bodily legible, for example by differentiating cues across the back and chest and, when feasible, using shear-like cutaneous signals to convey sustained drag. Impact cues can mark momentum changes at accelerations, dives and pull-ups, and sharp turns. While prior systems show that wind enhances immersion and flying sensation [16, 36], we specify flight-specific informational uses and bodily layouts for these cues.

Guide the Body Into a Stable Flight Posture. Design should guide users toward coordinated control across the scapular, back musculature and core, rather than relying on arm motion alone. Training tasks, posture prompts, and haptic guidance can help users form a stable body coordinate system and develop correction strategies. This operationalizes situated haptics by tying guidance to flight-specific action organization and bodily anchors [7].

5 Limitations and Future Work

Our sample was relatively homogeneous in cultural background and educational attainment. As an exploratory study, our findings are based on a single VR flying experience and self-reported accounts, and should be validated with prototypes and controlled comparisons to assess generalizability and efficacy.

Future work will recruit more diverse participants and examine a wider range of VR flying scenarios and control styles. We will build a wearable haptic prototype to instantiate and refine the proposed principles, and systematically manipulate key cues such as wind and drag, responsiveness, and impact cues. We also plan to compare different flying body models (e.g., wingsuit, jetpack, vehicle-based, and head-driven flight) to examine how the four-layer pathway varies across activities. We hope this four-layer perspective and the associated principles can serve as shared language for designing and evaluating future VR flying systems.

References

- [1] David M Blei, Andrew Y Ng, and Michael I Jordan. 2003. Latent dirichlet allocation. *Journal of machine Learning research* 3, Jan (2003), 993–1022.
- [2] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [3] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative research in sport, exercise and health* 11, 4 (2019), 589–597.
- [4] Grigore C Burdea. 2000. Haptics issues in virtual environments. In *Proceedings Computer Graphics International 2000*. IEEE, IEEE Computer Society, Geneva, Switzerland, 295–302.
- [5] Changyeol Choi, Joohee Jun, Jiwoong Heo, and Kwanguk Kim. 2019. Effects of virtual-avatar motion-synchrony levels on full-body interaction. In *Proceedings of the 34th ACM/SIGAPP symposium on applied computing*. ACM, Association for Computing Machinery, Limassol, Cyprus, 701–708.
- [6] Jean-Baptiste Chossat, Daniel K. Y. Chen, Yong-Lae Park, and Peter B. Shull. 2019. Soft Wearable Skin-Stretch Device for Haptic Feedback Using Twisted and Coiled Polymer Actuators. *IEEE Transactions on Haptics* 12, 4 (2019), 521–532. doi:10.1109/TOH.2019.2943154
- [7] Heather Culbertson, Samuel B Schorr, and Allison M Okamura. 2018. Haptics: The present and future of artificial touch sensation. *Annual review of control, robotics, and autonomous systems* 1, 1 (2018), 385–409.
- [8] Frédérique De Vignemont. 2010. Body schema and body image—Pros and cons. *Neuropsychologia* 48, 3 (2010), 669–680.
- [9] Benoit P. Delhay, Katie H. Long, and Sliman J. Bensmaia. 2018. Neural Basis of Touch and Proprioception in Primate Cortex. *Comprehensive Physiology* 8, 4 (Sept. 2018), 1575–1602. doi:10.1002/cphy.c170033
- [10] Huong Q Dinh, Neff Walker, Larry F Hodges, Chang Song, and Akira Kobayashi. 1999. Evaluating the importance of multi-sensory input on memory and the sense of presence in virtual environments. In *Proceedings IEEE virtual reality (cat. No. 99CB36316)*. IEEE, IEEE Computer Society, Houston, TX, USA, 222–228.
- [11] H Henrik Ehrsson, Charles Spence, and Richard E Passingham. 2004. That's my hand! Activity in premotor cortex reflects feeling of ownership of a limb. *Science* 305, 5685 (2004), 875–877.
- [12] David Gall, Daniel Roth, Jean-Philippe Stauffert, Jan Zarges, and Marc Erich Latoschik. 2021. Embodiment in Virtual Reality Intensifies Emotional Responses to Virtual Stimuli. *Frontiers in Psychology* 12 (2021), 674179. doi:10.3389/fpsyg.2021.674179
- [13] Shaun Gallagher. 2006. *How the body shapes the mind*. Clarendon press, Oxford, United Kingdom.
- [14] Mar Gonzalez-Franco and Tabitha C Peck. 2018. Avatar embodiment. towards a standardized questionnaire. *Frontiers in Robotics and AI* 5 (2018), 74.
- [15] Sebastian Günther, Florian Müller, Dominik Schön, Omar Elmoghazy, Max Mühlhäuser, and Martin Schmitz. 2020. Terminator: Understanding the interdependency of visual and on-body thermal feedback in virtual reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, Honolulu, HI, USA, 1–14.
- [16] Ping-Hsuan Han, Yang-Sheng Chen, Kong-Chang Lee, Hao-Cheng Wang, Chiao-En Hsieh, Jui-Chun Hsiao, Chien-Hsing Chou, and Yi-Ping Hung. 2018. Haptic around: multiple tactile sensations for immersive environment and interaction in virtual reality. In *Proceedings of the 24th ACM symposium on virtual reality software and technology*. Association for Computing Machinery, Tokyo, Japan, 1–10.
- [17] Ping-Hsuan Han, Tzu-Hua Wang, and Chien-Hsing Chou. 2023. Groundflow: Liquid-based haptics for simulating fluid on the ground in virtual reality. *IEEE*

- Transactions on Visualization and Computer Graphics* 29, 5 (2023), 2670–2679.
- [18] Ronan Hinchet, Velko Vechev, Herbert Shea, and Otmar Hilliges. 2018. Dextres: Wearable haptic feedback for grasping in vr via a thin form-factor electrostatic brake. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology*. Association for Computing Machinery, Berlin, Germany, 901–912.
 - [19] Ali Israr and Ivan Poupyrev. 2011. Tactile brush: drawing on skin with a tactile grid display. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, Vancouver, BC, Canada, 2019–2028.
 - [20] Hamed Jelodar, Yongli Wang, Chi Yuan, Xia Feng, Xiahui Jiang, Yanchao Li, and Liang Zhao. 2019. Latent Dirichlet allocation (LDA) and topic modeling: models, applications, a survey. *Multimedia tools and applications* 78, 11 (2019), 15169–15211.
 - [21] Andreas Kalckert and H Henrik Ehrsson. 2012. Moving a rubber hand that feels like your own: a dissociation of ownership and agency. *Frontiers in human neuroscience* 6 (2012), 40.
 - [22] Konstantina Kiltani, Raphaela Groten, and Mel Slater. 2012. The Sense of Embodiment in Virtual Reality. *Presence* 21, 4 (Nov. 2012), 373–387. doi:10.1162/PRES_a_00124
 - [23] Fengguan Li, Jiahong Chen, Guanpeng Ye, Siwei Dong, Zishu Gao, and Yitong Zhou. 2023. Soft robotic glove with sensing and force feedback for rehabilitation in virtual reality. *Biomimetics* 8, 1 (2023), 83.
 - [24] Pinyao Liu, Ekaterina R Stepanova, Alexandra Kitson, Thecla Schiphorst, and Bernhard E Riecke. 2022. Virtual transcendent dream: empowering people through embodied flying in virtual reality. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New Orleans, LA, USA, 1–18.
 - [25] Matthew R Longo, Friederike Schüür, Marjolein PM Kammers, Manos Tsakiris, and Patrick Haggard. 2008. What is embodiment? A psychometric approach. *Cognition* 107, 3 (2008), 978–998.
 - [26] Ryan P McMahan, Doug A Bowman, David J Zielinski, and Rachael B Brady. 2012. Evaluating display fidelity and interaction fidelity in a virtual reality game. *IEEE transactions on visualization and computer graphics* 18, 4 (2012), 626–633.
 - [27] Tim Moesgen, Ramyah Gowrishankar, and Yu Xiao. 2024. Designing Beyond Hot and Cold—Exploring Full-Body Heat Experiences in Sauna. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction*. Association for Computing Machinery, Cork, Ireland, 1–14.
 - [28] Hye Sung Moon, Grady Orr, and Myoungsoon Jeon. 2023. Hand tracking with vibrotactile feedback enhanced presence, engagement, usability, and performance in a virtual reality rhythm game. *International Journal of Human–Computer Interaction* 39, 14 (2023), 2840–2851.
 - [29] Charles M Oman. 1982. A heuristic mathematical model for the dynamics of sensory conflict and motion sickness. *Acta Oto-Laryngologica* 94, sup392 (1982), 4–44.
 - [30] Roshan Lalintha Peiris, Wei Peng, Zikun Chen, Liwei Chan, and Kouta Minamizawa. 2017. Thermovr: Exploring integrated thermal haptic feedback with head mounted displays. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, Denver, CO, USA, 5452–5456.
 - [31] Domenico Prattichizzo, Francesco Chinello, Claudio Pacchierotti, and Monica Malvezzi. 2013. Towards wearability in fingertip haptics: a 3-dof wearable device for cutaneous force feedback. *IEEE Transactions on Haptics* 6, 4 (2013), 506–516.
 - [32] Lisa Rebenitsch and Charles Owen. 2016. Review on cybersickness in applications and visual displays. *Virtual reality* 20, 2 (2016), 101–125.
 - [33] Max Rheiner. 2014. Birdly an attempt to fly. In *ACM SIGGRAPH 2014 Emerging Technologies (SIGGRAPH '14)*. Association for Computing Machinery, New York, NY, USA, 1. doi:10.1145/2614066.2614101
 - [34] Bernhard E Riecke. 2011. Compelling self-motion through virtual environments without actual self-motion: using self-motion illusions (“vection”) to improve user experience in VR. *Virtual reality* 8, 1 (2011), 149–178.
 - [35] Maria V Sanchez-Vives and Mel Slater. 2005. From presence to consciousness through virtual reality. *Nature reviews neuroscience* 6, 4 (2005), 332–339.
 - [36] Kazuya Shimato, Yuta Goto, and Shogo Okamoto. 2024. Drone rider: Enhancing VR flying experiences via directed wind stimuli. In *2024 IEEE 13th Global Conference on Consumer Electronics (GCCE)*. IEEE, IEEE, Kokura, Kitakyushu, Japan, 735–737.
 - [37] Kazuya Shimato, Yuta Goto, and Shogo Okamoto. 2024. Drone Rider: Foot vibration stimuli to enhance immersion and flight sensation in VR. *Applied Sciences* 14, 24 (2024), 12019.
 - [38] Mel Slater. 2009. Place illusion and plausibility can lead to realistic behaviour in immersive virtual environments. *Philosophical Transactions of the Royal Society B: Biological Sciences* 364, 1535 (2009), 3549–3557.
 - [39] Mel Slater, Daniel Pérez Marcos, Henrik Ehrsson, and Maria V. Sanchez-Vives. 2009. Inducing illusory ownership of a virtual body. *Frontiers in Neuroscience* 3, 2 (Sept. 2009), 214–220. doi:10.3389/neuro.01.029.2009 Publisher: Frontiers.
 - [40] Mel Slater, Bernhard Spanlang, Maria V Sanchez-Vives, and Olaf Blanke. 2010. First person experience of body transfer in virtual reality. *PloS one* 5, 5 (2010), e10564.
 - [41] Kay M Stanney, Ronald R Mourant, and Robert S Kennedy. 1998. Human factors issues in virtual environments: A review of the literature. *Presence* 7, 4 (1998), 327–351.
 - [42] Mengshi Yang and Ruochen Hu. 2025. “Overlapping Our Worlds”: Designing Biometric-Based Haptic Interactions to Enhance Synchrony in Long-Distance Relationships. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, Yokohama, Japan, 1–8.
 - [43] Jun Zhang, Han Li, Aiguo Song, Yizhuang Ding, and Juan Wu. 2022. Thermal perception for information transmission: Theoretical analysis, device design, and experimental verification. *IEEE Transactions on Haptics* 15, 4 (2022), 679–692.
 - [44] Yaying Zhang, Bernhard E Riecke, Thecla Schiphorst, and Carman Neustaedter. 2019. Perch to fly: Embodied virtual reality flying locomotion with a flexible perching stance. In *Proceedings of the 2019 on Designing Interactive Systems Conference*. Association for Computing Machinery, San Diego, CA, USA, 253–264.

A Study Materials

This appendix summarizes the study materials used for data collection. Participants first completed a body-map elicitation task (Figure 4) to externalize perceived bodily engagement and sensations during VR flight. They then took part in a semi-structured interview, guided by the questions listed below.

A.1 Body Map Elicitation Method

Participants annotated the front and back body outlines using three visual encodings: dots to mark control points or locations of force exertion, arrows to indicate directions of body movement or postural adjustment, and colored patches to highlight sensory regions. The number and placement of annotations were unconstrained. Participants provided brief verbal explanations of their markings during the task.

A.2 Semi-structured Interview Guide

The following interview questions were used to guide the post-deployment reflection:

- (1) **What moments during the VR flying experience stood out to you most, and why?**
(Probe for concrete events and triggers: takeoff/landing, sudden turns, acceleration, collisions, near-ground passing; ask what they saw/felt in the body and what made it memorable.)
- (2) **How did the experience of flying differ from your real-life or expected flying experience?**
(Probe for expectation sources: movies / games / airplane travel / dreams; ask what felt missing or exaggerated—speed, control effort, wind, weightlessness, fear / excitement, viewpoint height, or body involvement.)
- (3) **Were there any moments when the experience felt especially “real” or especially “unreal”? What triggered that feeling?**
(Probe for realism cues: visual flow, latency, control responsiveness, audio, wind/airflow, vibration; ask whether mismatches between motion and bodily sensation made it feel unreal, e.g., strong visual acceleration without corresponding bodily cues.)
- (4) **Which parts of your body felt the most “active” or “involved” during the flight?**
(Probe for active control vs passive sensation: arms / shoulders / back / core / legs / head / eyes; ask what they were doing, such as tensing, balancing, stabilizing, steering, bracing, breathing changes.)

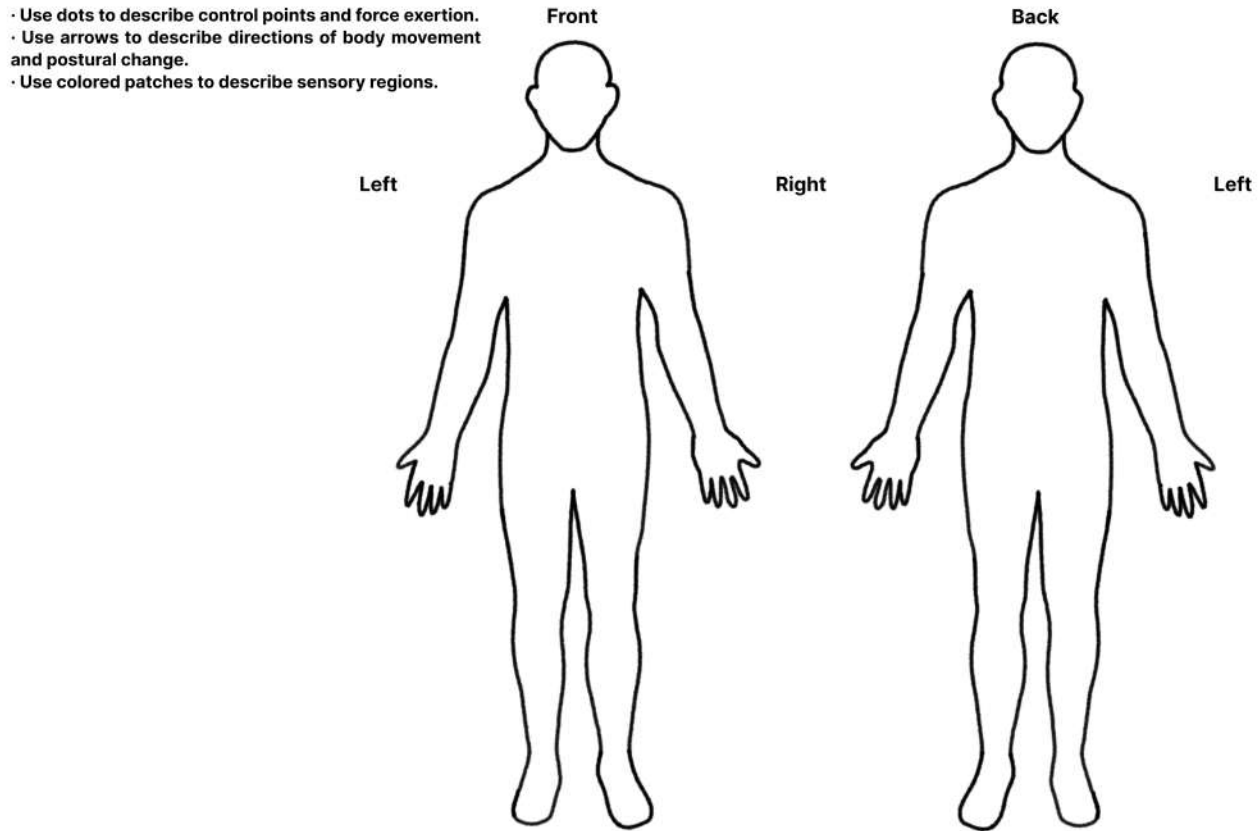


Figure 4: Body map template used for eliciting participants' bodily sensations during VR flying

- (5) **Were there any body parts you felt should have been involved but were not?**
(Probe for "expected" flight mappings: wings / back / shoulder blades, torso expansion, leg steering, hand / arm flapping; ask what involvement they expected and why, and whether the lack reduced control, presence, or comfort.)
- (6) **What kind of impact did the environment have on your actions during the flight (e.g., wind, gravity, buoyancy, or body floating)?**
(Probe for perceived forces and compensations: resisting wind, leaning into turns, bracing under acceleration, "floating" sensations; ask which environmental cue was most informative for speed/direction and how it changed their posture or control strategy.)
- (7) **If you could change one thing to make the flying feel more natural for your body, what would you change?**
(Probe for designable elements: control mapping, responsiveness, viewpoint motion, added haptic cues (wind / vibration / pressure), reducing discomfort; ask for one prioritized change and expected effect.)
- (8) **What kinds of bodily cues or feedback would help you feel more in control during flight (e.g., on the skin, muscles, joints, or balance)?**
(Probe for cue types and locations: airflow on face/arms, vibration intensity patterns, skin-stretch/pressure on torso, proprioceptive cues; ask what each cue would signal—speed, acceleration, turning, altitude, collision—and where it should be felt.)
- (9) **Imagine an "ideal" VR flying experience for you: what should your body be doing, and what should the system be doing to support that?**
(Probe for an end-to-end scenario: takeoff, cruising, maneuvering, landing; ask for body posture/movement, control division across body parts, and system support (guidance, constraints, feedback timing) that makes the experience feel coherent and safe.)